

L06: PCA

C. V. Jawahar

IIIT Hyderabad

January 27, 2026

Recap:

- Problems of interest:
 - Learn a function $y = f(\mathbf{W}, \mathbf{x})$ from the data.
 - (a) Classification (b) Regression
 - Learn Feature Transformations $\mathbf{x}' = \mathbf{W}\mathbf{x}$ or $\mathbf{x}' = f(\mathbf{W}, \mathbf{x})$
 - (a) Feature Normalization (b) PCA (today)
- Algorithms/Approaches:
 - Nearest Neighbour Algorithm
 - Linear Classification: $\text{sign}(\mathbf{w}^T \mathbf{x})$
 - Decide as ω_1 if $P(\omega_1 | \mathbf{x}) \geq P(\omega_2 | \mathbf{x})$ else ω_2 .
 - Linear Regression: (a) closed form and (b) GD
- Supervised Learning:
 - Notion of Training, Validation and Testing
 - Performance Metrics
 - Notion of Loss Function, (eg. MSE), Regularization.
 - Role of Optimization, Convex and non-Convex optimization
 - Closed form solution, Gradient Descent, Eigen vector solns.

PCA: Principal component Analysis

A classical, popular dimensionality reduction technique.

- Unsupervised
- Linear
- ① PCA: Dimensions that preserve maximum variance
 - $\mathbf{x}' = \mathbf{W}\mathbf{x}$
 - Problem: How to find $\mathbf{W}$?
- ② PCA as Compression
 - Dimensionality reduction that allow minimal loss in the data.
- ③ Eigen Faces
 - A powerful application of PCA
 - Face representation and compression.

Recap: Optimization problems with EVec as Solutions

Problem: Maximize $\mathbf{w}^T \mathbf{A} \mathbf{w}$ such that $\mathbf{w}^T \mathbf{w} = 1$ (or $\|\mathbf{w}\| = 1$)

We form an objective with the help of a lagrangian (λ) as

$$J(\mathbf{w}, \lambda) = \mathbf{w}^T \mathbf{A} \mathbf{w} - \lambda(\mathbf{w}^T \mathbf{w} - 1)$$

Differentiating wrt $\mathbf{w}$ and equating to zero leads to:

$$\mathbf{A} \mathbf{w} = \lambda \mathbf{w}$$

Soln: $\mathbf{w}$ is the eigen vector corresponding to the largest eigen value.

- 1 If we are given N points $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ in d dimension (say $d = 2$), what will be a good representing point $\mathbf{p}$? (hint: optimize the sum of square distance to all the points!)
- 2 If we are given N points $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ in d dimension (say $d=1$), what will be a good representing line (in 2D) (or hyper plane in general) $\mathbf{p}$? (hint: optimize the sum of square distance to all the points!)
- 3 What happens when we approximate a point by projecting to a line? What is the approximation error? (If we approximate a d -dimensional data in d' dimension, (where $d' < d$) what is the error of approximation or dimensionality reduction?

Fitting a line that minimize Orthogonal Distances¹

We want to find $\mathbf{w}$ that define the line and norm 1.0. Let us consider that all samples are mean subtracted.

$$\min \sum_{i=1}^N (\mathbf{x}_i^T \mathbf{x}_i - (\mathbf{w}^T \mathbf{x}_i)^2)$$

Alternatively,

$$\max \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i)^2 \text{ subject to: } \mathbf{w}^T \mathbf{w} = 1$$

$$\max \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} \text{ subject to: } \mathbf{w}^T \mathbf{w} = 1$$

Solution to this is the eigen vector corresponding to the largest eigen value of $\mathbf{X}^T \mathbf{X}$ or $\mathbf{\Sigma}$

¹<https://www.youtube.com/watch?v=likhONUdvnc>

Appreciating PCA

Maximum Variance Direction: 1st PC a vector $\mathbf{v}$ such that projection on to this vector capture maximum variance in the data (out of all possible one dimensional projections)

$$\frac{1}{n} \sum_{i=1}^n (\mathbf{v}^T \mathbf{x}_i)^2 = \mathbf{v}^T \mathbf{X} \mathbf{X}^T \mathbf{v}$$

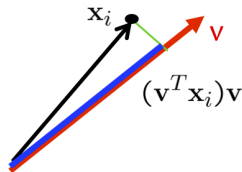
Minimum Reconstruction Error: 1st PC a vector $\mathbf{v}$ such that projection on to this vector yields minimum MSE reconstruction

$$\frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - (\mathbf{v}^T \mathbf{x}_i) \mathbf{v}\|^2$$

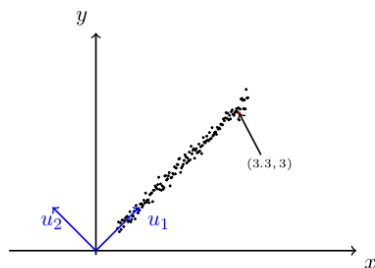
$$\text{blue}^2 + \text{green}^2 = \text{black}^2$$

black² is fixed (it's just the data)

So, maximizing blue² is equivalent to minimizing green²



Reconstruction: Numerical Example



- $u_1 = [1, 1]$ and $u_2 = [-1, 1]$ are the new basis vectors
- Let us convert them to unit vectors
 $u_1 = \left[\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right]$ & $u_2 = \left[\frac{-1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right]$

- Consider the point $x = [3.3, 3]$ in the original data
- $\alpha_1 = x^T u_1 = 6.3/\sqrt{2}$
 $\alpha_2 = x^T u_2 = -0.3/\sqrt{2}$
- the perfect reconstruction of x is given by (using $n = 2$ dimensions)

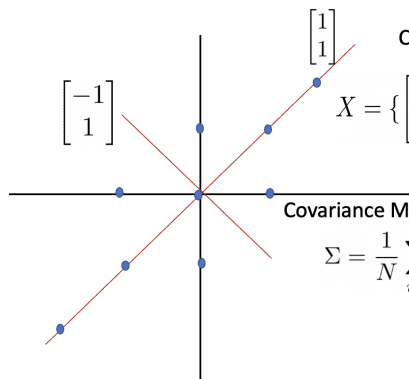
$$x = \alpha_1 u_1 + \alpha_2 u_2 = [3.3 \quad 3]$$

- But we are going to reconstruct it using fewer (only $k = 1 < n$ dimensions, ignoring the low variance u_2 dimension)

$$\hat{x} = \alpha_1 u_1 = [3.15 \quad 3.15]$$

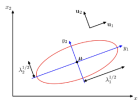
(reconstruction with minimum error)

Worked Out Example - I



Note: Eigen vectors are not normalized for simplicity

Consider nine samples in 2D



$$X = \left\{ \begin{bmatrix} 2 \\ 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} \begin{bmatrix} -2 \\ -2 \end{bmatrix} \right\}$$

Covariance Matrix:

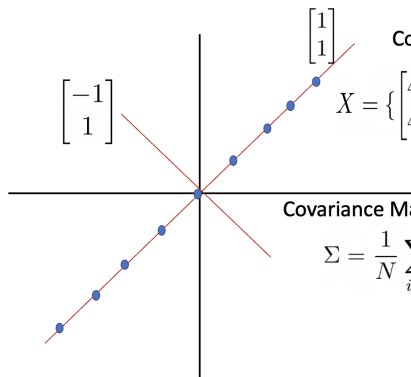
$$\Sigma = \frac{1}{N} \sum_{i=1}^N [x_i - \mu][x_i - \mu]^T$$

$$\Sigma = \frac{1}{9} \begin{bmatrix} 12 & 10 \\ 10 & 12 \end{bmatrix}$$

Eigen vectors: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Eigen values: $\lambda_1 = 22$ and $\lambda_2 = 2$

Worked Out Example - II



Note: Eigen vectors are not normalized for simplicity

Consider nine samples in 2D

$$X = \left\{ \begin{bmatrix} 4 \\ 4 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} \begin{bmatrix} -2 \\ -2 \end{bmatrix} \begin{bmatrix} -3 \\ -3 \end{bmatrix} \begin{bmatrix} -4 \\ -4 \end{bmatrix} \right\}$$

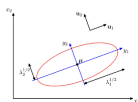
Covariance Matrix:

$$\Sigma = \frac{1}{N} \sum_{i=1}^N [x_i - \mu][x_i - \mu]^T$$

$$\Sigma = \frac{1}{9} \begin{bmatrix} 60 & 60 \\ 60 & 60 \end{bmatrix}$$

Eigen vectors: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Eigen values: $\lambda_1 = 120$ and $\lambda_2 = 0$



PCA: Algorithm

1. Input: N samples of D dimension
2. Compute Covariance Matrix
3. Compute Eigen Values and Eigen Vectors of Covariance Matrix
4. Select 'd' eigen vectors corresponding to 'd' largest eigen values
5. Arrange them as rows of 'A' (a 'd' X 'D' matrix)
6. Find lower dimensional representations as: $x' = Ax$

$$\Sigma = \frac{1}{N} \sum_{i=1}^N [x_i - \mu][x_i - \mu]^T$$

How much to Reduce?

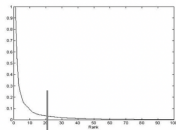
How many Eigen Values?

How many eigen values are required so that “most” of the information in the covariance/data is preserved? Consider the popular eigen expansion:

$$\begin{aligned}\Sigma &= \sum_{i=1}^D \lambda_i v_i v_i^T \\ &= \lambda_1 v_1 v_1^T + \lambda_2 v_2 v_2^T + \cdots + \lambda_{N-1} v_{N-1} v_{N-1}^T + \lambda_N v_N v_N^T\end{aligned}$$

Eigen values (Lambda) are in decreasing order. Even if we discard the small eigen values, most information is preserved.

Typical Eigen
Value Spectrum



$$d = \min_p \frac{\sum_{i=1}^p \lambda_i}{\sum_{i=1}^D \lambda_i} \geq 0.95$$

18

- All our faces look very different. But are they so different? Consider all our frontal faces are of size 100×100 or of size 10^4 .
 - What will be the mean face?
 - What about eigen values, rank of Σ
 - What about eigen vectors?
- Is it possible to develop a representation for human faces? (for solving tasks like face recognition)
 - That is compact (much smaller than 10^4)
 - Different for different faces.
 - How to formulate it as $\mathbf{x}' = \mathbf{W}\mathbf{x}$

Faces and Eigen Vectors



Representation and Reconstruction of Face from 16 EVs



Example

Eigen Faces



62 X 47 Image D = 2914; N = 1348

Vectors displayed as 2D Image

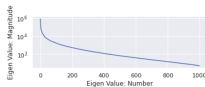
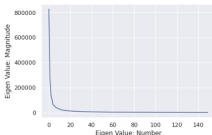


Mean
2914 X 1



Covariance Matrix is
2914 X 2914

First 16 Eigen Vectors
out of 2914



Log-scale

<https://towardsdatascience.com/eigenfaces-recovering-humans-from-ghosts-17606c328184>

How many coeff. are required to represent a block?

144 to 60, 16, 6, 3



Figure: Original, Blocks of size 12 X 12 in 60, 16, 6 and 3

Problem 1

Consider images of size 100×100 and we have 200 such images. Assume means are subtracted.

- ① What is the size of the covariance matrix?
- ② What is the rank of the covariance matrix? (guess!)
- ③ What is the size of XX^T and X^TX and what are their ranks?
- ④ How are the Eigen values of X^TX and XX^T related?
- ⑤ How are the Eigen vectors of X^TX and XX^T related?

Problem 2

Consider images of size 100×100 ($O(10^4)$) and we have 200 ($O(10^2)$) such images. We know that computing eigen vectors is a costly operation. How do we compute **W**?

Problem - 4

Matrix Completion Problem: Can we guess/compute/complete the missing elements of the matrix:

$$\begin{bmatrix} ? & 1 & 7 & ? & ? \\ ? & ? & ? & 8 & ? \\ 18 & ? & ? & 12 & 6 \\ ? & ? & ? & ? & 2 \\ ? & ? & 21 & 6 & ? \end{bmatrix}$$

if we know that this is a rank-1 matrix (or every row is a multiple of each other)²

Consider the following problem: Explain the notations and what we try to find. (Q: What is this summation over? What is the min over?)

$$\min \sum_i \sum_j (A_{ij} - B_{ij})^2 \text{ s.t. } \text{rank}(B) = 1$$

²Read later: <https://web.stanford.edu/class/cs168/l/l9.pdf>

Problem - 5

Continuing the numerical example we saw in the class with eigen values and eigen vectors. Consider three points $\mathbf{x}_1 = [5.0, 5.0]^T$, $\mathbf{x}_2 = [3.0, 3.3]^T$, $\mathbf{x}_3 = [-3.0, 0.0]^T$

- Find the low dimensional ($d = 1$) representations $\mathbf{x}' = \mathbf{W}\mathbf{x}$ What is $\mathbf{W}$?
- Find the reconstructed point and show the “error” magnitude in this new representation

- It is common to represent signal/data using some orthogonal basis. Discrete Fourier Transform (DFT) is an example that use complex exponential basis.
- Discrete Cosine Transform(DCT) (a close friend of DFT) is popular in the compression standards such as JPEG. We retain only a small number of coeff; rest as zero; thus obtain compression.
- The optimal linear transform to do this is popularly known as KL Transform (where eigen vectors of the covariance form the basis). Our PCA is a close relative.

Why is PCA/KL-Transform based compression algorithms not yet popular in the standards we all use?

Discussions Point - III

Consider a two dimensional representation with each dimensions as x_1 and x_2 . i.e., $\mathbf{x} = [x_1, x_2]^T$.

We know that we need to do a feature normalization to obtain

$$\mathbf{x}' = [x'_1, x'_2]^T$$

A $x'_i = \frac{x_i}{\alpha_i}$

B $x'_i = x_i - \alpha_i$ (after the class?)

How do we represent

- 1 $\mathbf{x}' = \mathbf{L}\mathbf{x}$? What is $\mathbf{L}$
- 2 What should be the matrix $\mathbf{A}$ such that the Euclidean distance in $\mathbf{x}'$ is same as Mahalanobis distance in $\mathbf{x}$. i.e.,

$$[\mathbf{x} - \mathbf{y}]^T \mathbf{A}^{-1} [\mathbf{x} - \mathbf{y}] = [\mathbf{x}' - \mathbf{y}']^T [\mathbf{x}' - \mathbf{y}']$$

Problem-II

Q: Consider a linear transformation $d \rightarrow d$ (i.e., $\mathbf{W}$ is a square matrix)

$$\mathbf{x}' = \mathbf{W}\mathbf{x}$$

We use a K-NN algorithm (the same K and distance as Euclidean distance) in original and new space.

- Will the performance (say accuracy) of the algorithm be same in both the space for any $\mathbf{W}$? i.e., with $\mathbf{x}$ and $\mathbf{x}'$? (Discuss)
- If no, what should be the condition on $\mathbf{W}$ to guarantee that?
- If $\mathbf{W}$ is null (all elements zero), what happens? If $|\mathbf{W}| = 0$, what happens?

Review Question - II (one, none or more correct)

$$\frac{TP}{P}$$

is known as:

(a) Accuracy (b) Precision (c) Recall (d) None of the above

Review Question - IV (one, none or more correct)

Two SMAI students (Raju and Sheela) worked on the same problem with the same measurements/features and samples, except that their feature orderings were different. (i.e., $\mathbf{x}$ and $\mathbf{x}'$ were permutations.) Identify correct statement(s).

- (a) Both got the same accuracy with KNN (same K and Eucli. distance)
- (b) Both got different accuracy with KNN (same K and Eucli. distance)
- (c) Their confusion matrices were different i.e., elements (cells) were swapped.
- (d) Both had the same Covariance Matrices (Hint: $\Sigma = \frac{1}{N} \sum_{i=1}^N [\mathbf{x}_i - \mu][\mathbf{x}_i - \mu]^T$)
- (e) Both had covariance matrices of the same Rank.
- (f) Both had covariance matrices where cells (elements) were swapped.

What Next:?

- Application of SVD and Eigen Decomposition
- More Insights into Supervised Learning
- Bayesian View and Optimal Classification
- Practical Issues in Optimization

Discussions Point -I

Consider a situation when we continue to get one sample at a time. We have mean (μ_N) and variance (σ_N^2) computed and available at sample N .

Now we get the $N + 1$ sample. How do we compute the new mean? Ans:

$$Ans : \mu_{N+1} = \frac{\mu_N \times N + x_{N+1}}{N + 1}$$

How do we compute σ_{N+1}^2 ?

Where do we need such “online” computations?

Review Question - II (one, none or more correct)

A certain test for disease is known to have True positive of 0.6 and False Positive of 0.1.

A population of 100 people (where 60 of them are infected) undergoes this test.

What could be the confusion matrix?

- (a) $\begin{bmatrix} 0.6 & 0.4 \\ 0.1 & 0.9 \end{bmatrix}$ (b) $\begin{bmatrix} 0.6 & 0.4 \\ 0.9 & 0.1 \end{bmatrix}$ (c) $\begin{bmatrix} 0.6 & 0.2 \\ 0.1 & 0.3 \end{bmatrix}$ (d) $\begin{bmatrix} 0.58 & 0.42 \\ 0.15 & 0.85 \end{bmatrix}$
(e) None of the above

Discussions Point - I

In supervised learning, we use the training data $\mathcal{D}_{Tr}$ to develop a solution and then evaluate it on the test data $\mathcal{D}_{Te}$

Q: Why is that a solution optimized only on training data as

$$\min_{\mathbf{W}} \mathcal{L}(\mathbf{W}, \mathbf{x}) \quad \mathbf{x} \in \mathcal{D}_{Tr}$$

works reasonably well on the test data also?

- ① We believe in luck.
- ② Once it is trained on $\mathcal{D}_{Tr}$, it then works well on any data.
- ③ Actually, we use $\mathcal{D}_{Te}$ also for training. Eqn. is incorrect.
- ④ What we care is actually the performance on the $\mathcal{D}_{Tr}$; not $\mathcal{D}_{Te}$.
- ⑤ It is some magic. No explanation found till today.

Review Questions: Let us submit in the first 10 mins

- ① Numerically computing rank of a 3×3 matrix
- ② The system of linear equations $Ax = b$ has?
- ③ Suppose a disease is prevalent in 1% of the population. Its medical diagnosis is 90% accurate in both directions. Given that a person tested positive, what is the chance, he actually has the disease (rounded to nearest integer)?
- ④ We know that the optimal classifier for two equally probable (equal Prior probability) classes (days of months) $N(23, \sigma^2)$ and $N(33, \sigma^2)$ is 28.
If the variance of the second class becomes double, then the the optimial classification threshold will increase or decrease?
- ⑤ A man is known to speak truth 2 out of 3 times. He throws a die and reports that number obtained is a four. Find the probability that the number obtained is actually a four.

Problem 3

- We know the $d \times N$ data matrix $\mathbf{X}$ with every column as data elements. Show that $\mathbf{A} = \mathbf{X}\mathbf{X}^T$ is Symmetric. Show that $\mathbf{A}$ is PSD.³
- If $\mathbf{A}$ is real symmetric PSD matrix, we know $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, then what is $\mathbf{V}^T\mathbf{A}\mathbf{V}$? What is this process called?

³A matrix A is PSD if $z^T A z \geq 0 \quad \forall z \in R^d$.